

Lab 1: ESP32 Microprocessor and I/O

Objectives:

1. To learn how to do simple signal conditioning and read digital inputs into the processor.

Tasks:

1. Read a digital input
2. Interface a digital signal to ESP32-S3 (ESP) through an open-collector circuit
3. Build and test a Schmitt trigger input debounce
4. Implement interrupts

References:

[YouTube] [EEVBlog - Schmitt Trigger Tutorial](#)

[Github] [Example Code](#)

Pre-Lab

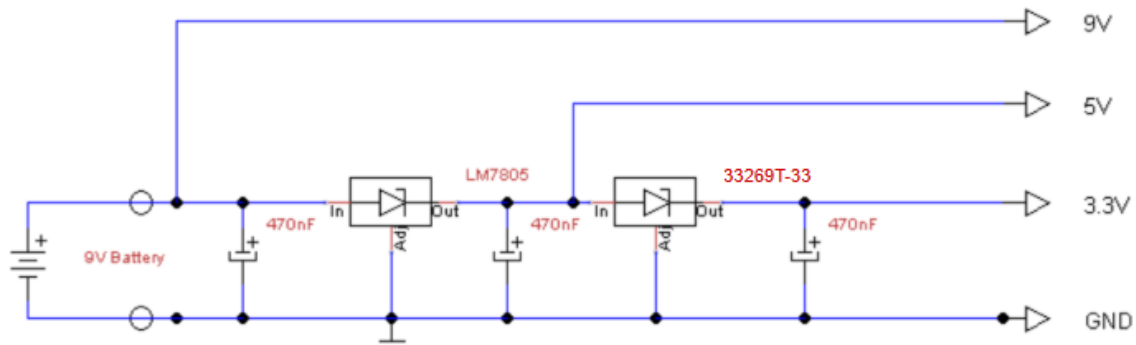
1. Set-up your ESP so that you can upload code and use the OLED LCD (Lab 0)
2. Familiarise yourself with the data sheet for [TCRT5000](#), [LM311](#), [7805](#), [33269T](#)

Note: Keep detailed records of the code you've written, your circuits, what worked and what didn't.

Lab 1 instructions

1. Set-up the ESP32 power supply

- Using a 7805 and 7833 regulator and capacitors, build a 5V / 3.3V regulator to power the ESP. Confirm voltages with your scope or voltmeter before you connect anything to them. Do not power your esp32 from both a 3.3V supply and 5V supply at the same time (they might fight)!
- **NOTE: Do not power the esp32 from both it's 3.3Volt pin and the USB port! For the same reason as above. If you want to program the esp32 either disconnect the external power to the 3.3volt pin, or power the esp only from the 5V pins**
- Ensure you are able to power the ESP without the usb connector.



2. Reading a simple digital input.

- Connect a micro switch to a digital input (internally pulled up) and print its state on the OLED display by polling input using `digitalRead()`
- Read and display the switch state using interrupt and an interrupt service routine (switch bounce will cause erroneous readings at this stage)

3. Switch de-bouncing

- Build a Schmitt trigger using an LM311 comparator, and low pass filter the switch output to de-bounce the signal (see lecture notes)
- Using two channels on an oscilloscope verify that upstream of the Schmitt trigger the signal has multiple transitions between high and low and that downstream of the Schmitt Trigger the signal toggles in a single transition to the new state.
- Verify that reading the switch with interrupts is now working correctly.

4. Read a tape sensor with a digital input (optional but helpful)

- Build a circuit to interface a TCRT5000 reflectance sensor to the ESP by passing through a 311 comparator. Use a potentiometer to allow you to adjust the sensitivity of the circuit. Confirm that the ESP can now detect black tape on white paper.